SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: CSA1610
COURSE NAME: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

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QUESTIONS: 05 **Total Marks:50**

Annexure A
MOCK INTERVIEW

Criteria	Technical Knowledge *(2.5)*	Problem Solving *(2.5)*	Analytical Thinking *(1.5)*	***Communication(1.5)***	Confidence *(1)*	Response to Questions(1)	***Total(10)***
Weightage	*25%*	25%	15%	15%	10%	*10%*	*100%*
Q1							
Q2							

Code of Conduct:

I Tejaswini.M(192411079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Tejaswini.M

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Annexure A

Mock Interview – Sample Questions (5 Questions)

Q1. Dataset: Retail Transactions

TID	Items
T1	Milk, Bread
T2	Milk, Butter
T3	Bread, Butter
T4	Milk, Bread, Butter
T5	Milk, Eggs

Question:
Based on the dataset, explain how association rule mining can improve sales in a retail store. Provide a practical example.

Association Rule Mining is a data mining technique used to discover relationships between items in a dataset. In the given retail dataset, Milk and Bread occur together in T1 and T4. Therefore, the store can create a rule such as **Milk → Bread**. Based on this pattern, the store can place Milk and Bread together or offer them as a combination, which can increase sales.

Q2. Dataset: Transaction Data

TID	Items
T1	A, B
T2	B, C
T3	A, C
T4	A, B, C
T5	B, C

Question:
Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used.

Apriori and FP-Growth are algorithms used to find frequent itemsets. **Apriori** generates candidate itemsets and repeatedly scans the database to find their support. **FP-Growth** uses an **FP-Tree** and avoids generating a large number of candidate itemsets. Apriori is suitable for smaller datasets because it is simple to understand, while FP-Growth is more suitable for large datasets because it is more efficient.

Q3. Dataset: Transaction Data

TID	Items
T1	A, B
T2	A
T3	B
T4	A, B
T5	A

Question:
Explain support and confidence using the dataset and describe their importance in association rule mining.

Support measures how frequently an itemset occurs in the complete dataset. In the given dataset, A and B occur together in T1 and T4, so:

Support(A → B) = 2/5 = 40%

Confidence measures how often B occurs when A occurs. A occurs in T1, T2, T4 and T5, while A and B occur together in T1 and T4.

Confidence(A → B) = 2/4 = 50%

Support helps identify frequent patterns, while confidence helps determine how reliable an association rule is.

Q4 . Dataset: Multilevel Data

TID	Items
T1	Electronics, Mobile
T2	Electronics, Laptop
T3	Electronics, Mobile
T4	Electronics, Mobile
T5	Electronics, Laptop

Question:
Explain multilevel association rules using the dataset and compare them with multidimensional

Multilevel association rules discover relationships at different levels of detail in a hierarchy. For example, **Electronics → Mobile** represents a relationship between a general category and a specific item. **Multidimensional association rules** use different attributes or dimensions, such as customer age, location and product. Therefore, multilevel focuses on **different levels**, while multidimensional focuses on **different attributes**.

Q5. Dataset: Large Transaction Sample

TID	Items
T1	A, B, C
T2	A, B
T3	B, C
T4	A, C
T5	A, B, C

Question:
Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues.

Mining large datasets is challenging because there can be a huge number of transactions and item combinations. This increases processing time and memory requirements. Scalable methods address these problems using efficient data structures, reducing unnecessary candidate generation, and using parallel or distributed processing. **FP-Growth** is an example of a scalable method because it uses an FP-Tree to efficiently mine frequent patterns.